Predicting Patient Drug Ratings from Reviews

A Deep Learning and Generative AI Approach

UCI ML Drug Review Dataset — Multiclass Rating Prediction

Master's Mentorship Assignment

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1. introduction

The analysis of patient-generated drug reviews offers a promising avenue for post-market drug surveillance; monitoring real-world drug experience at scale, in a domain where structured clinical data is scarce and patient-written text fills the gap. Online review platforms such as Drugs.com aggregate hundreds of thousands of patient accounts describing drug effectiveness, side effects, and overall satisfaction in everyday, informal language. Automatically predicting the sentiment and rating behind these reviews could support clinical decision-making, therapy recommendation systems, and pharmacovigilance efforts, motivating a growing body of research at the intersection of natural language processing and healthcare.

This report documents a project undertaken as a mentorship assignment under Prof. Kobi Gal, designed to explore deep learning and generative AI approaches to predicting patient drug ratings from free-text reviews on the UCI ML Drug Review Dataset. The assignment was scoped as a focused exercise — expected to take a few hours — but grew substantially in scope as the work deepened: what began as a straightforward modeling task evolved into a four-day project spanning data cleaning, exploratory analysis, two modeling pipelines, explainability analysis, and a generative AI component, all conducted under meaningful compute constraints that shaped every methodological decision along the way.

The core tasks were: (1) predicting the patient's rating on a 10-class scale (1–10) and a coarser 3-class bucketing (≤4, 4–7, ≥7) from the review text, and (2) generating a concise summary and binary sentiment label for each review using a large language model. The dataset presents several inherent challenges; a heavily imbalanced U-shaped rating distribution, informal and nontechnical patient language, and genuine ambiguity in middle-range ratings, that made clean, conclusive results difficult to achieve within the available time and compute budget. What this report offers instead is an honest account of the process: the architectural choices made and why, the constraints encountered and how they shaped the methodology, and what the results can and cannot support as conclusions.

Through this project I practiced building models, reading and grounding decisions in research, and writing a comprehensive report, perhaps more comprehensive than needed. The most important thing I learned was the importance of proper resource management and conducting an experiment in a planned and methodical way. I am grateful to Prof. Gal for the opportunity he gave me.

The report is structured as follows: Chapter 2 describes the methodology; Chapter 3 covers data cleaning and EDA; Chapters 4 and 5 present the 10-class and 3-class models respectively; Chapter 6 describes the generative AI component; Chapter 7 compares our results to the original paper published with this dataset; and Chapter 8 consolidates limitations and next steps.

2. Methodology

2.1 Data and Splits

All modeling was performed on the UCI ML Drug Review Dataset, a publicly available collection of patient-written drug reviews scraped from Drugs.com. The dataset arrives pre-split into a training set (161,297 rows) and a test set (53,766 rows), each containing seven fields: a row identifier, drug name, condition being treated, free-text review, patient rating (1–10), date posted, and a "useful count" reflecting how many other users found the review helpful.

The training set was further divided into a train split (85%) and a validation split (15%) using stratified sampling; preserving the original rating distribution in both halves, with **random_state=42** fixed throughout for reproducibility. The test set was held out entirely and not used during any model development, hyperparameter tuning, or architecture decision. All reported performance metrics come from the validation set unless explicitly stated otherwise.

The **usefulCount** column was excluded from all models as a potential input feature. This value only accrues after a review is posted, using it as a predictor would mean training on information that would not exist at inference time for a genuinely new review, a form of data leakage.

2.2 Evaluation Metrics

Given the heavily imbalanced rating distribution identified during EDA (rating 10 alone accounts for $\sim 31.6\%$ of training examples, while several middle ratings each account for under 5%), plain accuracy was rejected as the primary metric. A model could score deceptively well on accuracy simply by defaulting to the majority class. We instead use macro-F1 as the primary metric throughout; it computes F1 separately for each class and averages them equally, giving minority classes the same weight as majority classes in the final score.

Results are additionally reported with:

- Full classification report (per-class precision, recall, F1, support)
- Confusion matrix (full 10×10 for the 10-class task; 3×3 for the 3-class task)
- ±1 accuracy: the proportion of predictions falling within one rating point of the true value, capturing near-miss performance that a hard-correctness metric would obscure
- Statistical significance tests when comparing two models on the same validation set

2.3 Class Weighting

To counteract class imbalance during training, class weights were computed from the training split only (not the full training set, to avoid leaking validation-set distribution into training decisions) using $sklearn's\ compute\_class\_weight('balanced', \ldots)$, and passed to the loss function as explicit weights. This causes the model to penalize mistakes on minority classes more heavily than on majority classes, discouraging the model from simply defaulting to high-frequency ratings.

2.4 Architecture Selection Framework

Rather than selecting a model architecture by intuition or convention, we grounded the choice in a systematic comparative study (Lu, Ehwerhemuepha & Rakovski, 2022) that evaluated seven deep learning architectures for free-text classification under class imbalance, the closest existing benchmark to our task. The architecture covered four broad families:

Sequence models (RNN family): RNN, GRU, LSTM, and Bi-LSTM; all designed to process text sequentially, token by token, capturing dependencies across the sequence. These models have strong inductive biases for language but can struggle with long-range dependencies and are inherently sequential, limiting parallelization during training.

Convolutional model: CNN — applies fixed-size filters across the token sequence, capturing local n-gram patterns efficiently. Faster to train than RNN-family models and less sensitive to sequence length, but unable to capture long-range dependencies as naturally.

Transformer encoder: A self-attention-based encoder trained from scratch; captures global dependencies across the full sequence in parallel, without the sequential bottleneck of RNN-family models, but requires substantially more data and compute to train well from scratch due to lacking the structural inductive biases that help RNNs and CNNs generalize with less data.

Pretrained BERT: A Transformer encoder pretrained on large general-purpose corpora (Wikipedia, BookCorpus) and fine-tuned on the target task; carries broad prior language understanding into the task, making it more data-efficient than a from-scratch Transformer encoder.

The study found the Transformer-based family outperformed the RNN and CNN families in most scenarios. CNN emerged as the closest competitor to Transformers among the lighter architectures, particularly when class distributions were more balanced, a finding that directly informed our model choice for the 3-class task in Chapter 5.

Within the pretrained BERT family, we selected DistilBERT (Sanh, Debut, Chaumond & Wolf, 2019) rather than full BERT-Base. DistilBERT is a distilled version of BERT trained via knowledge distillation, retaining approximately 97% of BERT's language understanding performance at 60% of the training time and 40% fewer parameters. This tradeoff was confirmed on a task directly comparable to ours, Yelp review star-rating prediction, where DistilBERT scored only ~0.5% below full BERT-Base while training nearly twice as fast. Given our project's compute constraints (free-tier Colab GPU access, rationed usage quota), this made DistilBERT the practical and well-justified choice over full BERT.

Beyond using DistilBERT as a standalone fine-tuned classifier (Model A), we also explored a hybrid architecture (Model B) in which the fine-tuned DistilBERT is frozen and used purely as a text feature extractor, producing a fixed 768-dimensional representation that is concatenated with separately learned embeddings for drugName (32-dim) and condition (16-dim), and passed to a small classification head trained from scratch. This dramatically reduces the number of trainable parameters, making each epoch roughly 12 times faster than full fine-tuning.

2.5 Generative AI Component

For the generative AI component, we selected **Qwen3-4B** (Alibaba, 2025), a 4-billion parameter instruction-tuned language model loaded in float16 precision. Four prompting approaches were tested, progressing from a simple zero-shot instruction to increasingly rich few-shot prompts grounded in real dataset examples with clinical role framing. The two best-performing approaches were each run on a stratified 500-row sample of the test set, producing two Excel output files with identical structure differing only in the prompting approach used.

3. Data Cleaning and Exploratory Data Analysis

3.1 Data Cleaning

The UCI ML Drug Review dataset arrived as a pre-split train set (161,297 rows) and test set (53,766 rows), each with seven columns: a row ID, drug name, condition, free-text review, rating (1–10), date, and a "useful count" of how many users found the review helpful. Both files loaded cleanly with no row loss.

Several data quality issues were identified and addressed:

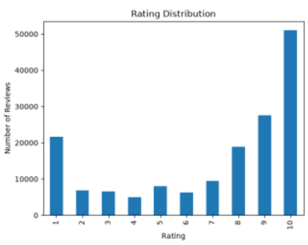
- **Missing condition values.** 899 rows had no condition listed. *Fix:* imputed as "Unknown".
- **Scraping artifacts in condition.** Several rows contained leftover HTML helpfulness-count snippets (e.g., "95 users found this comment helpful.") instead of an actual condition. 900 such rows in train and 271 in test. *Fix:* detected via pattern matching, relabeled as "Unknown" alongside the true missing values, rather than dropped.
- **HTML entities in review text.** Apostrophes, quotes, and similar characters appeared as raw HTML entities (e.g., ', ", &). *Fix:* unescaped using Python's html module.
- **Stray quote marks in review text.** Leftover formatting artifact from the original file's quoting. *Fix:* stripped from every review.
- **Single-character reviews.** Inspecting the shortest reviews revealed a small number reduced to one character (e.g., "I", "-", "G"), carrying no usable signal for predicting rating. *Not fixed*: kept in the dataset rather than dropped, noted as a known source of low-quality signal to consider when interpreting model performance.

The cleaned train and test sets were saved as train_clean.csv and test_clean.csv.

3.2 Exploratory Data Analysis

EDA was scoped specifically to inform modeling decisions, not as general data exploration. Every finding below maps directly to a concrete choice made later in the project.

Rating distribution is heavily imbalanced. Ratings follow a U-shape: people tend to leave a review when they feel strongly, so extremes dominate while middle ratings (2–6) are comparatively rare. The same imbalance carries into the 3-class bucketing (Class 1: 24.8%, Class 2: 8.9%, Class 3: 66.3%).



Decision*:* use class weighting in training, and report macro-F1 alongside accuracy, since accuracy alone would reward a model that simply defaults to majority classes.

Review length is tightly clustered, with a thin tail of outliers. The 90th, 95th, and 99th percentiles (141, 146, 154 words) sit close together, while the maximum (1,894 words) is a rare extreme. Checking the rating distribution of long reviews (>160 words, 0.4% of the data) against the full dataset showed they skew toward extreme ratings rather than being noise, people write more when they feel strongly.

Decision: set $max\_sequence\_length = 160$, truncating rather than dropping long reviews, applied only at tokenization time so the stored review text and downstream GenAI summarization are unaffected.

Naive tokenization inflated the vocabulary. An initial whitespace-based word count produced **154,505 "unique words",** but 74.5% contained punctuation (e.g., "good." and "good" counted separately). Re-tokenizing with a regex that strips punctuation while preserving contractions (don't, i'm) **reduced** the true vocabulary to **46,898 words**, of which 18,581 appear 5 or more times.

Decision: cap the embedding vocabulary at this 18,581-word threshold for any from-scratch embedding model, mapping rarer words to a single unknown token.

Condition has a systemic truncation issue. Checking Condition and Drug-Name fields for anomalies found drug-Name free of issues, but condition values were frequently truncated; missing characters at the start, end, or both (e.g., "Bipolar Disorde" for "Bipolar Disorder"), affecting one of the dataset's most common conditions and ~9.3% of rows overall. Since condition isn't used as a primary model input, this does not affect model performance and is documented as a known data limitation.

4. The 10-Class Model

4.1 Architecture Selection

To choose a model, we searched for relevant research rather than relying on intuition. We found a study conducted in a hospital setting, evaluating which architectures perform best on free-text classification under class imbalance. The best-performing model family was Transformer-based (encoder architectures), followed by RNN- and CNN-type models. Notably, in that same study, full BERT scored poorly, for two reasons: it was pretrained on general text (Wikipedia, books) rather than clinical language, and the medical notes in that study averaged 557 words, exceeding BERT's 512-token input limit and truncating most documents.

This distinction mattered for our decision: training a Transformer encoder from scratch requires a large amount of data and high computational cost, since it lacks the built-in structural assumptions that RNNs/CNNs have. We chose BERT (specifically, its distilled form) **despite** its poor showing in that study, because neither of the two reasons for its poor performance applies to our case; drug reviews contains conversational language rather than clinical jargon, and our reviews are far shorter (95th percentile of 146 words) than BERT's 512-token limit.

A second study we reviewed, on Yelp review star-rating prediction, compared different versions of BERT. From this, we chose **DistilBERT** over full BERT-Base, since it is meaningfully smaller and faster while the accuracy tradeoff is small (~0.5% lower accuracy, ~60% of the training time).

4.2 Training Environment and Struggles

Earlier project stages (data cleaning, EDA) were done locally in VS Code. However, fine-tuning DistilBERT, while lighter than other Transformer-based models, still carries meaningful computational cost, so this notebook was run in Google Colab using its free GPU.

Model A (text-only) used only the review column for classification. Training began with 4 epochs (~12 minutes each); validation macro-F1 was still clearly improving and showed no sign of convergence, so we continued training for 5 additional epochs with an early-stopping mechanism in place (set to stop after just 1 epoch with no improvement, given the high per-epoch cost). This mechanism never triggered, every one of the 5 additional epochs improved on the last, so training simply ran its full epoch budget. Macro-F1 continued improving steadily across all 9 epochs. with no clear sign of approaching a ceiling. We recognize that this situation, being limited by time and compute rather than training to genuine convergence, limits how strong our conclusions about the model's *true* ceiling can be. What we report should be read as a performance floor: the model had not finished learning when we stopped. We hope to revisit deeper training in the future if time and resources allow.

4.3 Model B: Metadata Embeddings on Frozen DistilBERT

Model B extends Model A by treating the fine-tuned DistilBERT as a frozen text encoder, concatenating its 768-dimensional output with separately learned embeddings for drugName and condition, and training a small classification head on top.

Because only the classification head required training, each epoch took roughly one minute, dramatically faster than Model A, and the model reached genuine convergence with early stopping triggering naturally. Despite producing similar headline numbers to Model A, formal statistical tests (McNemar's test, sign test, and Wilcoxon signed-rank test) confirmed the improvement is real and statistically significant.

Model B was selected as the final 10-class model.

	Model A (text-only)	**Model B (text + drugName + condition)**
Accuracy	0.58	0.59
Macro-F1	0.5511	0.5777
Within ±1 rating	83.1%	82.4–83.1%*

**Varied slightly across separate training runs of Model B's head, reflecting normal training randomness rather than a meaningful difference.*

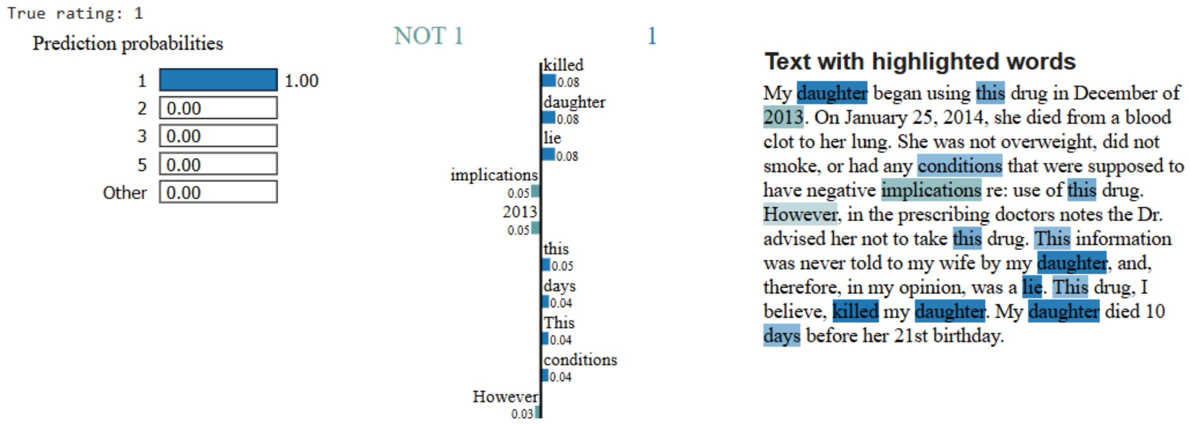
4.4 Explainability

Explainability visualizations were generated in a separate session, after the trained model parameters were saved to disk. By this point, Colab's free-tier GPU quota had been exhausted, forcing a fallback to Colab's CPU. Running on CPU rather than GPU meant the explainability algorithms had to be limited to a **sample of 100 reviews**, rather than the full validation set. This is a real limitation, results from 100 reviews are noisier and less representative than results computed across the entire dataset, but the available time and compute made this the practical choice, and the resulting visualizations still offer a meaningful, if approximate, picture of the model's behavior.

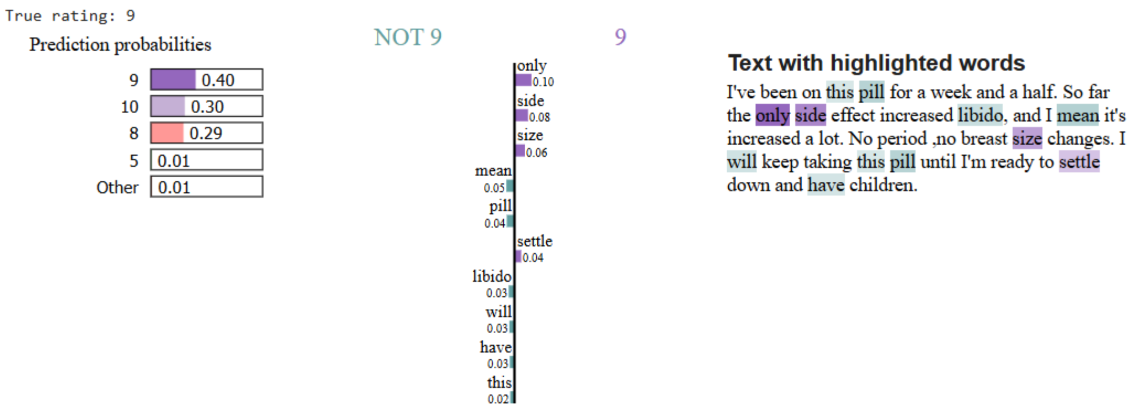
We used two complementary explainability techniques:

- **LIME** (Local Interpretable Model-agnostic Explanations), to examine word-level contributions to individual predictions; applied consistently across both Model A and Model B, allowing direct comparison of which words drove each model's reasoning on the same reviews.

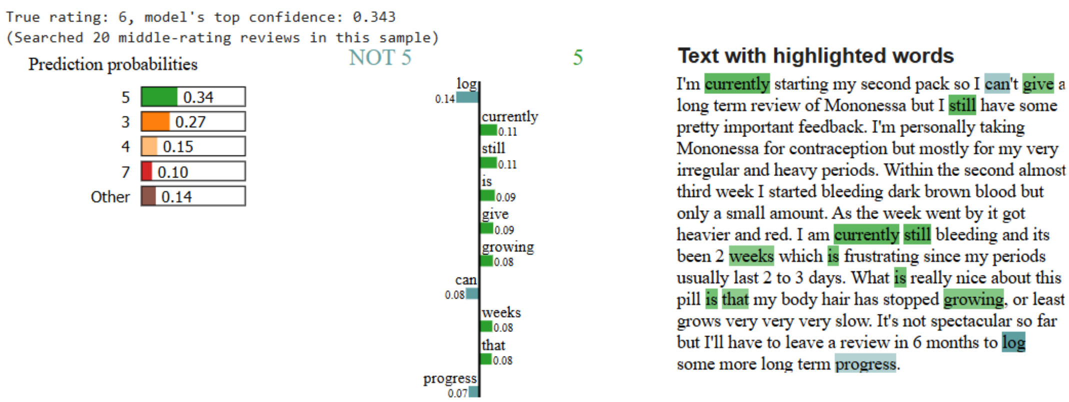
- **A custom global word-direction analysis** (leave-one-word-out), to identify which words, on average across many reviews, push predictions toward high or low ratings, giving a dataset-wide view that complements LIME's single-review focus.



For rating 1, words like "killed," "daughter," and "lie", reflecting a severe adverse outcome, dominate the prediction with near-total confidence (1.00).

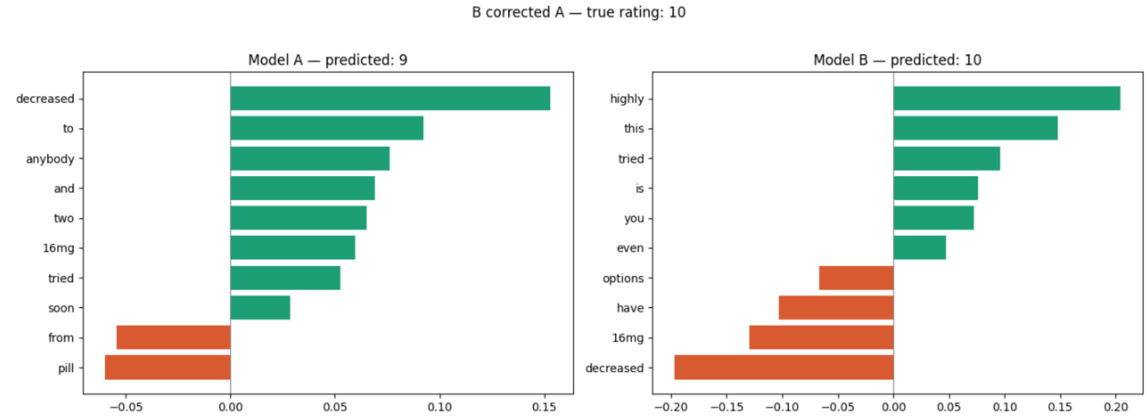


At the high end, rating 9's prediction is comparatively uncertain (0.40, with 0.30 for 10 and 0.29 for 8 close behind); the review's language ("only side effect," "increased libido") is mildly positive but unremarkable.



Searching specifically for the model's least confident predictions among middle-range ratings (4–7) surfaces genuinely ambiguous cases. For a true rating of 6, the model's top prediction (5) reaches only 0.34 confidence, closely contested by 3, 4, and 7, far less certain than its near-100% confidence at the extremes. This tracks the review's genuinely mixed language ("still bleeding," "frustrating," but also "really nice," "stopped growing"), showing the model's uncertainty reflects real ambiguity rather than noise.

```
=== B corrected A ===
True: 10, Model A predicted: 9, Model B predicted: 10
Drug: Buprenorphine, Condition: Opiate Dependence
```



A direct review-by-review comparison on this sample found Model B corrected 9 of Model A's mistakes while introducing only 5 new ones, consistent with its statistically significant overall improvement.

4.5 Conclusion

Most Errors Are Near-Misses

Across both models, most mistakes are not random; they fall close to the true rating. 83.1% of Model A's predictions (and 82.4% of Model B's) land within ± 1 of the true rating. Looking at the confusion matrix, errors consistently cluster around the diagonal: a true rating of 9 is most often confused with 8 or 10, not with 1 or 2. This indicates the model reliably captures the direction of sentiment, even when it misses the exact intensity.

```
[[2520  324  102   53   97   22   17   41   19   48]
 [ 222  606   67   34   53    6   13   21    8   10]
 [ 120  143  471   66   83   19   29   28    8   10]
 [  72   31   50  444   61   18   32   26    5   13]
 [  98   49   71   69  679   61   62   77   18   18]
 [  38   18   40   53   93  495   73   94   27   20]
 [  36   17   29   25  106  123  713  205   95   69]
 [  65    9   26   16  137   59  314 1351  651  206]
 [  72   10    9   11   81   31  150  712 2258  796]
 [ 128   13   19    7   90   30  134  687 2139 4401]]
```

	precision	recall	f1-score	support
1	0.75	0.78	0.76	3243
2	0.50	0.58	0.54	1040
3	0.53	0.48	0.51	977
4	0.57	0.59	0.58	752
5	0.46	0.56	0.51	1202
6	0.57	0.52	0.55	951
7	0.46	0.50	0.48	1418
8	0.42	0.48	0.44	2834
9	0.43	0.55	0.48	4130
10	0.79	0.58	0.66	7648

True \ Predicted	Class 1 (≤4)	Class 2 (4–7)	Class 3 (≥7)
Class 1 (≤4)	5,325	359	328
Class 2 (4–7)	436	1,328	389
Class 3 (≥7)	492	657	14,881

Metric	Class 1	Class 2	Class 3
Precision	0.85	0.57	0.95
Recall	0.89	0.62	0.93
F1	0.87	0.59	0.94

Macro avg (simple average across the three classes, each weighted equally):

- **Precision:** 0.79, **Recall:** = 0.81, **F1:** = 0.80

Weighted avg (weighted by class support, so it reflects the actual class sizes):

- **Precision:** 0.89**, Recall:** 0.89**, F1:** 0.89

Re-bucketing Model A's existing predictions into the three target classes ; without training any new model, simply by reframing the existing 10-class predictions — produces a clear jump in performance: accuracy rises from 0.58 to 0.89 and macro-F1 from 0.55 to 0.80, since most near-miss errors (e.g., predicting 8 instead of 9) now fall within the same bucket. The gap between weighted-average F1 (0.89) and macro-average F1 (0.80) shows the model still performs noticeably worse on minority Class 2, mirroring the imbalance seen in the 10-class results, just less pronounced. This matrix was only computed for Model A; the equivalent for Model B should be calculated in future work. This re-bucketed result is the natural starting point for the 3-class task, explored next.

5. The 3-Class Model

5.1 Approach and Model Selection

As discussed in Section 4.2, this stage faced the same time and compute constraints as the 10-class task, by this point, the free Tesla T4 GPU quota had been exhausted, making it impractical to fine-tune DistilBERT a second time specifically for 3 classes. Instead, we used the existing fine-tuned DistilBERT (Model A from Chapter 4) as a baseline by simply re-bucketing its 10-class predictions into the three target buckets ($\leq 4, 4-7, \geq 7$). described in Section 4.5. The equivalent full-dataset analysis for Model B was not computed, for the same time constraints discussed throughout this report. It is reasonable to assume that a DistilBERT model *trained directly* on 3-class labels would perform even better, since it would only need to learn the two relevant decision boundaries (e.g., distinguishing 6 from 7) rather than all nine boundaries required by the 10-class formulation. This remains an open question for future work.

Given the GPU constraint, we used this opportunity productively: rather than simply reusing the existing model, we trained a new, lightweight architecture specifically for the 3-class task, allowing a real comparison between a heavyweight pretrained Transformer and a lightweight from-scratch alternative. We selected a **CNN**, based on the same medical-notes comparative study: that study found CNN to be the closest competitor to Transformer-based models, particularly when class distributions are more balanced; and our 3-class bucketing, while still imbalanced, is notably less skewed than the original 10-class distribution. CNN was also reported as dramatically faster to train than Transformer-based models, making it a practical fit given our remaining compute budget.

5.2 Training the CNN

Unlike DistilBERT, the CNN required building its own vocabulary and embedding layer from scratch; and our EDA-derived vocabulary decision (18,581 words, min_frequency=5) finally applied directly, unlike with DistilBERT, which uses its own fixed, pretrained WordPiece vocabulary that cannot be resized. The CNN was trained on the full training set with class weighting (the 3-class distribution remains imbalanced at roughly 66%/25%/9%), and evaluated using the same macro-F1, full classification report, and confusion matrix approach used for Model A.

Training was capped at 6 epochs ($\sim 10 - 12.5$ minutes each); roughly the same per-epoch time Model A took, despite the CNN being a far lighter architecture. The difference: Model A trained on Colab's Tesla T4 GPU, while the CNN trained on CPU only, since GPU quota had been exhausted by this point. This illustrates how much GPU acceleration matters; a much smaller model on CPU took about as long per epoch as a much larger model on GPU. As with Model A, the CNN had **not converged** when training stopped; validation macro-F1 was still climbing steadily across all 6 epochs (0.5959 → 0.6153 → 0.6199 → 0.6735 → 0.6911 → 0.7062), reaching a final macro-F1 of **0.7062** on the full validation set. What we *can* learn from this, despite the lack of convergence, is the model's performance floor and its general behavior pattern; and the trajectory itself (steady, still-accelerating improvement) is informative on its own, suggesting a meaningfully higher ceiling is achievable with more training time.

5.3 Comparison to the Baseline

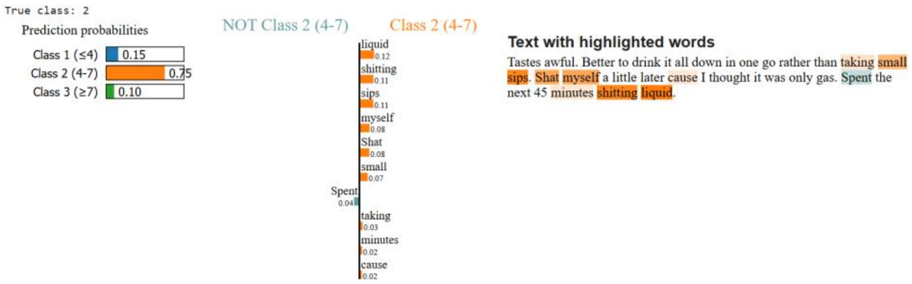
Comparing the CNN against the re-bucketed baseline from Chapter 4, the most solid comparison available is against **Model A's** full-dataset re-bucketed result (macro-F1 0.80, computed on all 24,195 validation rows) rather than Model B's, since Model B's re-bucketed performance was only ever computed on a 1,000-row sample (0.8155) and was never evaluated on the full set. On that same basis, the CNN's full-dataset macro-F1 (0.7062) falls clearly below Model A's (0.80), a meaningful gap, even accounting for the CNN's lack of convergence. The smaller, less reliable 1,000-row comparison against Model B (where the CNN scored 0.8297 vs Model B's 0.8155, with the difference not statistically significant per McNemar's test, $\chi^2 = 2.65, p = 0.104$) should be read with caution given the sample size, and is not used as the primary basis for comparison here.

Taken together, the evidence; full-dataset CNN performance (0.7062) well below Model A's full-dataset re-bucketed result (0.80), despite the small-sample CNN-vs-Model B test being a toss-up, supports selecting a DistilBERT-based model (re-bucketed) over the CNN for the 3-class task. We specifically select Model B over Model A for this purpose by inference rather than direct full-dataset measurement: Model B was shown to be statistically significantly better than Model A on the unbucketed 10-class task (Section 4.3), so it is reasonable to expect this advantage carries over once re-bucketed, though this has not been directly confirmed and remains a clear next step. We are explicit that this conclusion rests on an uneven comparison, different evaluation set sizes, neither model retrained natively for 3 classes, rather than a fully controlled experiment. Addressing this properly (training a dedicated 3-class DistilBERT and evaluating both models on the same full validation set) is identified as a clear next step, discussed further in the Limitations / Next Steps chapter.

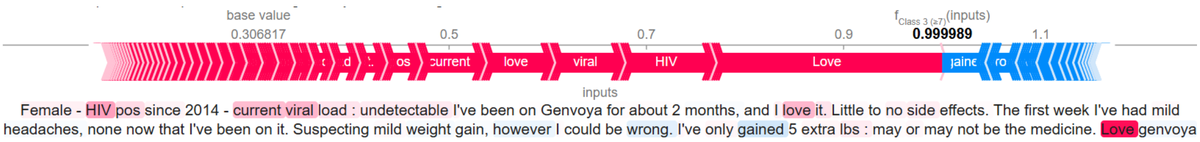
Although the CNN was not selected as the final model, we chose to perform explainability analysis on it rather than on the re-bucketed DistilBERT. The reasoning is twofold: the CNN was trained natively as a 3-class model, meaning its word-level reasoning is directly grounded in the 3-class label space rather than re-interpreted from 10-class predictions; and having already examined DistilBERT's behavior in Chapter 4, applying the same techniques to a different architecture here allows us to observe how a lighter, from-scratch model reasons about the same task; not as a comparison, but as an independent lens on what language-level signals the 3-class problem relies on.

5.4 Explainability

We applied LIME to the CNN across one review from each true class, to examine word-level reasoning across the rating spectrum:



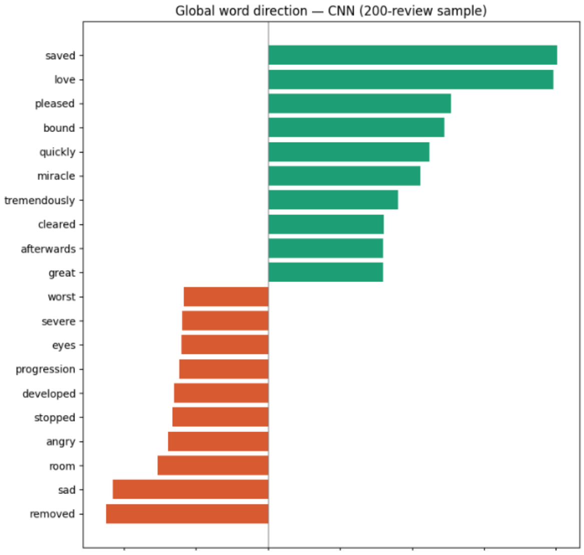
For a Class 2 (4–7) review, the prediction is more measured (0.75); words like "shitting," "sips," and "myself" reflect a blunt, complaint-heavy account of side effects, without the extreme finality seen in the Class 1 example.



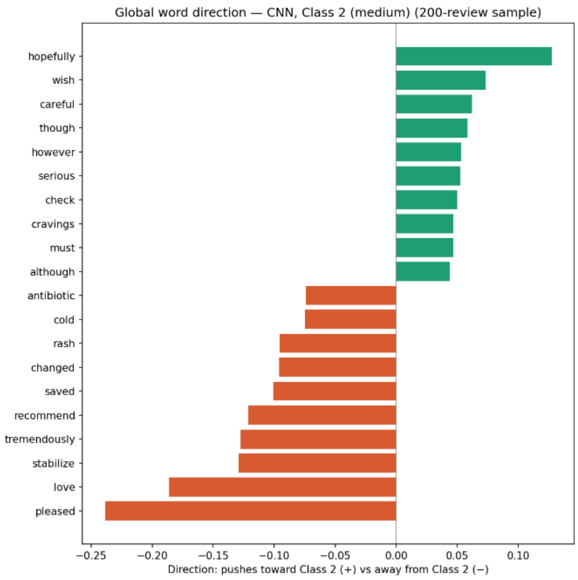
For a Class 3 (≥7) review, the model was fully confident (1.00). To confirm this finding isn't an artifact of one particular explainability method, we cross-validated it using SHAP; an independently-grounded approach based on game-theoretic attribution, rather than LIME's local surrogate modeling, on this same review, and found closely matching results: "current viral load," "HIV," and "Love" driving the prediction toward Class 3, with "gained" (referencing possible weight gain) pulling slightly against it. This agreement between two independently-grounded methods adds real confidence that the model's reasoning reflects genuine, relevant content rather than an artifact of either technique individually.

A larger 200-review sample was used here, versus 100 for DistilBERT in Chapter 4 (Section 4.4), since the CNN's much lower per-inference cost made a larger sample practical even on CPU.

Global word direction (leave-one-out). While LIME and SHAP each explain a single review, we also wanted a dataset-wide view of which words the model relies on most generally. We built a custom leave-one-out analysis: for each review in a 200-review sample, every word is individually removed and the resulting shift in predicted class probability is measured, then averaged across all reviews where that word appeared. This is, in our view, the strongest and most rigorous piece of explainability evidence in this section, since it isn't dependent on any single example and directly measures causal impact on the model's output, rather than approximating it through a surrogate model.



Across the full sample, words like "saved," "love," and "miracle" push most strongly toward Class 3, while "sad," "removed," and "angry" push toward Class 1; a clear, sensible separation that matches what a person would expect from sentiment-driven language.



Isolating the minority Class 2 specifically reveals a distinct vocabulary: words like "hopefully," "wish," and "careful" push toward Class 2, while strongly positive words ("love," "pleased," "recommend") push *away* from it. This makes sense , Class 2 represents a more uncertain, middling patient experience, characterized by hedging language rather than strong sentiment in either direction.

6. Generative AI Component

6.1 Model Selection

For the generative summarization and sentiment classification tasks, we selected **Qwen3-4B** (Alibaba, 2025); a 4-billion parameter instruction-tuned language model from the Qwen3 family. The selection was grounded in the model landscape analysis described in Section 2.5: Qwen3-4B sits at the lightweight end of capable open-source models, requiring approximately 8GB of GPU memory at float16 precision, making it deployable on a free-tier Tesla T4 GPU without quantization. We initially experimented with **Flan-T5-base** (250M parameters) as a lighter alternative, but found it insufficiently capable for the multi-task prompt format required here, struggling to produce both a structured summary and a sentiment label in a single generation call, regardless of prompting approach. Larger alternatives (Mistral-7B, Phi-4) would have offered stronger output quality but exceeded the available compute budget given the volume of rows to process.

A practical constraint worth documenting: this project required two separate Google Colab accounts to complete the generative AI component, since the free-tier GPU quota (approximately 5 hours per account) was exhausted during the classification modeling stages of Chapters 4 and 5 before this component could begin.

6.2 Task Design

For each review in the sample, Qwen3-4B was asked to produce two outputs:

1. **A summary of ≤10 words** capturing the key patient experience
2. **A sentiment classification** (positive or negative) based on the review text

Both outputs were generated by the generative model directly from the review text: sentiment was not derived from the predicted rating, keeping the GenAI component's output independent from the classification model's predictions. A separate column (rating_sentiment) was added to the output files as a cross-reference, derived from Model B's predicted rating (>5 = positive, ≤5 = negative), allowing readers to see where the GenAI-generated sentiment agrees or disagrees with the rating-based signal.

6.3 Prompt Engineering

We designed and tested four prompting approaches on a stratified 30-row sub-sample covering all 10 rating levels, evaluating each on format compliance (does it produce a clean Summary + Sentiment output?) and sentiment accuracy (does the sentiment match the predicted rating direction?).

Approach A — Zero-shot structured: A direct instruction with no examples, asking for both summary and sentiment in labeled format. Performance varied significantly across runs (70.0% in the first run, 83.3% in the second), suggesting sensitivity to sample composition rather than genuine robustness, the lowest accuracy in the first run and highest in the second. This instability led us to exclude it, while noting it warrants further investigation with a larger evaluation sample.

Approach B — Simple few-shot: Same structure as A, with two short generic examples (one clearly positive, one clearly negative) to anchor the expected output format. Achieved consistent results across both runs (76.7% and 80.0%), with 2-3 unknowns per run, reliable format compliance with modest accuracy.

Approach C — Rich few-shot with real examples: Same task framing as B, but with five real reviews from the actual dataset spanning ratings 1, 2, 5, 6, and 10 as examples. Achieved 73.3% and 76.7% across the two runs, consistently the third-ranked approach, suggesting that richer examples alone without additional role framing did not improve over simpler few-shot prompting.

Approach D — Clinically descriptive few-shot: Same five real examples as C, combined with a role framing ("You are a medical data analyst") and an explicit instruction to capture specific clinical detail rather than just sentiment direction. Achieved 76.7% and 80.0% across the two runs, matching B on accuracy while producing zero unknowns in both runs, the most consistent format compliance of any approach.

Selection: B and D were selected as the two final approaches; both were the most stable and highest-performing across both 30-row evaluation runs (76.7% and 80.0% respectively), unlike A which swung from lowest (70.0%) to highest (83.3%) across runs. A was additionally eliminated on qualitative grounds: its summaries were too generic to convey meaningful clinical information beyond what the sentiment label already captures. C was eliminated for consistently ranking third across both runs. D offered the additional advantage of zero unknowns across both runs and more clinically specific summaries. All conclusions are based on 30-row samples and should be treated as preliminary.

Example outputs (30-row sample, two rating levels):

Rating 5 — "Nexplanon, irregular bleeding since insertion..."

- A: "Nexplanon causes irregular bleeding." / Negative
- B: "Nexplanon causes irregular bleeding and long-term use." / negative
- C: "Irregular bleeding with no pregnancy concerns." / negative
- D: "Irregular bleeding patterns persist despite Nexplanon use." / negative

Rating 10 — "Aleve worked fast for gout swelling..."

- A: "Aleve effective for gout relief." / Positive
- B: "Effective for gout swelling with fast relief." / positive
- C: "Aleve effectively reduces gout swelling quickly." / positive
- D: "Aleve provided rapid relief for gout-related toe swelling." / positive

Looking at these examples, the qualitative progression from A to D is visible: A produces the shortest, most generic summaries; B adds modest specificity; C and D capture more clinical detail, with D consistently producing the most precise and informative summaries. All four approaches agree on sentiment across these three examples, with the differences emerging primarily in summary quality and specificity rather than classification accuracy.

6.4 Final Output Generation

Approaches B and D were run on a stratified 500-row sample of the test set (stratified by rating to preserve the dataset's natural distribution), generating summaries and sentiments for each review.

	Approach B	**Approach D**
Accuracy	77.6% (388/500)	77.2% (386/500)
Unknowns	57/500 (11.4%)	8/500 (1.6%)
Macro-F1	0.82	0.77
Agreement with each other	83.2%	

The two approaches achieve nearly identical accuracy, but differ substantially in format compliance: B failed to produce a parseable sentiment for 11.4% of rows versus only 1.6% for D. Their failure modes also differ: B tends to abstain rather than commit when uncertain (high precision 0.97, lower recall 0.72 on positive), while D tends to classify ambiguous reviews as negative rather than abstaining (high recall 0.95, lower precision 0.60 on negative).

Unknowns were not confined to the ambiguous middle class as expected, B's unknowns spread across Class 2 (15.6%) and Class 3 (13.7%), suggesting format compliance failures on clearly positive reviews too. D's unknown rate was negligible across all classes (0–2.2%). Two illustrative cases:

Case 1 — Only B unknown (rating 8, Contrave): A review describing initial side effects that resolved over time.

- B: "Contrave caused headaches, thirst, and constipation initially." / **unknown**
- D: "Headache, thirst, constipation resolved with miralax and increased dose." / **positive**

Case 2 — Both unknown (rating 9, Chantix): A review describing severe side effects alongside successful smoking cessation.

- B: "Effective smoking cessation with significant side effects." / **unknown**
- D: "Chantix caused severe nausea and vivid dreams but enabled 4-week smoke freedom." / **unknown**

Note that D's summary here exceeds the 10-word limit: a pattern confirmed across the full 500-row sample, where Approach D violated the word constraint in 13.4% of summaries (vs only 2.6% for B). The more descriptive prompting style that makes D's summaries richer also makes it harder for the model to stay within the word limit. Future work should enforce this constraint more strictly, for example by explicitly rejecting and regenerating any summary exceeding 10 words.

6.5 Discussion and Conclusions

Case 2 above illustrates a more fundamental issue: the binary positive/negative sentiment framing is a simplification that doesn't suit all drug reviews well. The assignment's 3-class bucketing arguably captures patient experience more honestly, and the GenAI model's uncertainty in these cases is less a failure than an honest signal that the task itself is harder than the binary framing suggests.

All conclusions below are tentative and require validation on larger samples. Approach B, given only generic extreme examples, classified confidently at the poles but struggled with complex reviews; adding borderline few-shot examples would be a natural next step. Approach D, given real patient reviews spanning the full rating range, handled nuanced language better (8 unknowns vs 57) but committed more readily in the wrong direction for borderline cases (lower precision). Neither approach is unambiguously better: they represent different points on the precision/recall/coverage tradeoff. Future work should test on larger samples, enrich B's few-shot examples with borderline cases, and consider whether a 3-class sentiment framing would better match the inherent structure of patient drug reviews.

7. Comparison to Gräßer et al. (2018)

7.1 Context and Motivation

Gräßer et al. (2018) is the paper published alongside the UCI ML Drug Review dataset and represents the closest direct baseline for this work: both studies use the same Drugs.com source data and the same 10-star rating system as the basis for classification. Understanding where our approach agrees with, extends, or diverges from their findings provides important context for interpreting our results.

7.2 Task and Methodology Differences

The two studies frame the problem differently in several important ways.

Task framing: Gräßer et al. performed aspect-based sentiment analysis; treating each review as containing three separable aspects (overall satisfaction, effectiveness, and side effects) and training separate classifiers for each. We treated the review holistically, predicting a single overall rating rather than decomposing it into aspects. Their approach is more granular and clinically informative; ours is simpler and more directly comparable to a standard text classification task.

Methods: Their classifiers were classical ML models (logistic regression, SVM) with bag-of-words and TF-IDF features; no deep learning, no pretrained embeddings. This is significant: they explicitly conclude by recommending deep learning as a natural next step, framing their work as a baseline to be surpassed. Our work can be read as a direct response to that recommendation, replacing their shallow feature representations with DistilBERT's contextual embeddings.

Use of condition: Both studies used condition as a variable, but in fundamentally different ways. Gräßer et al. used condition as a **domain**; training on one condition and testing on another to study transfer. We used condition as a **feature**; encoding it as a learned embedding concatenated with the text representation in Model B. Their finding that cross-domain transfer degrades performance significantly actually supports our design choice: if condition-specific language carries strong predictive signal (which their cross-domain results confirm), then explicitly encoding condition identity as a model input should improve performance, which is exactly what Model B's statistically significant improvement over Model A suggests.

Data split: They used a 75/25 train/test split; we used 85/15, giving our models slightly more training data.

7.3 Results Comparison

In-domain performance (Table 2): On the Drugs.com overall rating task, the most directly comparable to ours; Gräßer et al. achieved **92.24% accuracy and Cohen's kappa of 83.99** using a 3-class label derived from the 10-star rating. This is notably higher than our 3-class accuracy (0.89–0.91) and macro-F1 (0.80–0.84). However, this comparison requires important caveats: their 3-class thresholds may differ from ours (≤4, 4–7, ≥7), their evaluation used k-fold cross-validation on the full dataset rather than a held-out test set, and their result was on in-domain data without the class imbalance constraints we explicitly addressed through class weighting and macro-F1 reporting. Their Druglib.com results tell a different story; accuracy dropped to 69.88% / kappa 28.45 when evaluated on the secondary source, suggesting their model's strong Drugs.com performance does not generalize across data sources.

Cross-domain performance (Table 3): When training on one condition and testing on another, their accuracy dropped substantially; diagonal (in-domain) results ranged from 90–95%, while off-diagonal (cross-domain) results fell to 57–82%, with kappa scores dropping even more sharply (from ∼78 – 90 in-domain to 20–51 cross-domain). This confirms that condition-specific language is a meaningful source of signal that doesn't transfer freely, reinforcing the value of our Model B approach, which explicitly encodes condition identity rather than assuming the model will learn to generalize across it.

Error patterns: Both studies found that errors cluster at neighboring classes — their misclassification of mild/moderate side effects as adjacent categories mirrors our finding that 83.1–83.6% of our 10-class predictions fall within ±1 of the true rating. This convergence across very different methodologies suggests the near-miss error pattern is a property of the task itself rather than any particular model.

7.4 Key Shared Insights

Several findings and observations from Gräßer et al. proved directly relevant to our own design decisions and results, even though the two studies were conducted independently and with different methods.

Their observation that drug review language is informal, nontechnical, and domain-dependent; motivating their move away from lexicon-based approaches toward learned classifiers, also motivated our choice of pretrained contextual embeddings over bag-of-words. Our explainability findings reinforce this: the words driving our models' predictions ("saved," "miracle," "horrible," "bleeding") are everyday sentiment words rather than clinical terminology, consistent with their characterization of the language in this domain.

Their finding that patient activity is condition-dependent, with some conditions generating far more reviews than others, adds a condition-level dimension to the class imbalance problem we addressed at the rating level. Future work should account for both simultaneously.

Their framing of the task as pharmacovigilance; post-market drug surveillance where structured clinical data is scarce and patient-generated text fills the gap, provides useful context for interpreting the value of both studies. Both demonstrate that automated analysis of drug reviews is feasible and informative, even if not yet production-ready.

7.5 Summary

Gräßer et al. established a strong classical ML baseline on this dataset, particularly for in-domain Drugs.com classification. Our work extends their approach in three concrete directions: replacing bag-of-words with contextual pretrained embeddings (DistilBERT), explicitly encoding condition as a model feature rather than treating it as a domain variable, and adding a generative AI component for summarization and sentiment labeling. Their explicit recommendation of deep learning as a next step, and their finding that condition-specific language carries strong predictive signal, both directly motivated our architectural choices. The near-miss error pattern they observed has persisted into our deeper models, suggesting it reflects genuine ambiguity in how patients express middle-range satisfaction — a challenge that neither classical nor deep learning approaches have fully resolved.

8. Limitations and Next Steps

8.1 Data Limitations

The condition column suffers from a systemic truncation issue affecting ~9.3% of rows: values are frequently cut off at the start, end, or both (e.g., "Bipolar Disorde" for "Bipolar Disorder"). While this did not affect model performance since condition was used only as an embedding input rather than a primary feature, it represents a known data quality gap worth addressing in future work through fuzzy matching or manual correction of the most common truncated values.

Single-character reviews (e.g., "I", "-", "G") were retained in the dataset rather than dropped. These carry no usable signal for rating prediction and represent a minor source of low-quality training examples.

8.2 Training Constraints

The most pervasive limitation across this project was compute and time; specifically, reliance on free-tier Google Colab GPU access (Tesla T4), which is rationed and was exhausted multiple times, requiring two separate Colab accounts to complete the project. This constraint manifested in several concrete ways:

Neither the 10-class nor the 3-class model reached convergence. Model A's macro-F1 was still improving steadily across all 9 training epochs when training stopped; the CNN's macro-F1 was still climbing across all 6 epochs. Both reported results should be read as performance floors rather than ceilings, the models had not finished learning when training was halted.

Explainability sample sizes were reduced. The global word-direction analysis was limited to 100 reviews for DistilBERT and 200 reviews for the CNN (the CNN's lower per-inference cost making a larger sample practical on CPU). Results from these sample sizes are meaningful but noisier than a full-dataset analysis would produce.

Local execution was not possible. A metadata-detection mismatch in the transformers library prevented DistilBERT from running locally, forcing all Transformer-based work into Colab and making the project entirely dependent on cloud compute availability.

8.3 Modeling Gaps

No dedicated 3-class DistilBERT model was trained. The 3-class baseline was produced by re-bucketing Model A's 10-class predictions rather than training a model natively on 3-class labels. A dedicated 3-class DistilBERT would only need to learn two decision boundaries rather than nine, and would likely outperform the re-bucketed baseline, but this was not feasible within the available compute budget.

Model B's re-bucketed 3-class performance was only evaluated on a 1,000-row sample, never on the full validation set. The comparison between the CNN and Model B in Chapter 5 therefore rests on unequal evaluation conditions and cannot be treated as a fully controlled experiment.

The re-bucketed confusion matrix was only computed for Model A, not Model B. Future work should compute this for Model B to confirm whether its statistically significant 10-class advantage carries over to the 3-class setting.

8.4 GenAI Limitations

The GenAI evaluation was based on a 500-row sample of the test set rather than the full 53,766 rows, due to generation speed constraints (~1.2 seconds per review on T4). All reported accuracy and macro-F1 figures for the GenAI component should be treated as estimates rather than definitive results.

Prompting experiments were conducted on two 30-row samples — too small for statistically robust conclusions, as evidenced by Approach A's accuracy swinging from 70.0% to 83.3% across runs. Due to compute constraints, prompt tuning was limited to three experimental runs in total, falling well short of the iterative refinement prompt engineering typically requires. Future work should evaluate larger samples with dedicated tuning iterations per approach**.**

8.5 Next Steps

Taken together, the most impactful next steps in priority order are:

1. **Train a dedicated 3-class DistilBERT** and evaluate it alongside the CNN on the full validation set, this would replace the current uneven comparison with a properly controlled experiment.
2. **Continue training both models to convergence** — with more GPU access, running Model A and the CNN for 20+ epochs would establish true performance ceilings rather than floors.
3. **Compute Model B's full re-bucketed confusion matrix** to confirm whether its 10-class advantage carries over to 3 classes.
4. **Run the GenAI pipeline on the full test set** with a stricter word-count enforcement mechanism, and evaluate prompting approaches on a larger sample (500+ rows) for more reliable conclusions.

5. **Experiment with 3-class sentiment framing** in the GenAI prompts, and add borderline few-shot examples to Approach B to reduce unknowns on ambiguous reviews.
6. **Fix the condition truncation issue** via fuzzy string matching against a known drug condition vocabulary, improving the quality of the condition embedding input for Model B.
7. **Explore** prompting the generative model to produce aspect-specific summaries (benefits, side effects, overall) and adopt a 3-class sentiment framing (negative/neutral/positive); giving the model a middle-ground option for ambiguous reviews and improving alignment with the rating bucketing used throughout this project.

9. Conclusions

This project set out to predict patient drug ratings from free-text reviews using deep learning and generative AI, across two tasks: a 10-class rating prediction (1–10) and a coarser 3-class bucketing $(\leq 4, 4-7, \geq 7)$. The full pipeline — from data cleaning through modeling, explainability, and generative summarization — was completed under significant compute constraints, with all results representing performance floors rather than ceilings given that neither main model reached full convergence.

Data preparation identified and resolved five distinct data quality issues, with the most notable being scraping artifacts in the condition column and HTML entities in review text. EDA confirmed a heavily imbalanced U-shaped rating distribution, directly informing the use of class weighting and macro-F1 as the primary evaluation metric throughout.

10-class modeling compared two **DistilBERT**-based architectures. Model A (text-only fine-tuned DistilBERT) achieved macro-F1 of 0.5511 and 83.1% ±1 accuracy on the validation set. Model B, which added learned **embeddings** for **drugName** and **condition** on top of a frozen DistilBERT encoder, achieved macro-F1 of 0.5777; a statistically significant improvement confirmed across three independent tests (McNemar's, sign test, Wilcoxon). Model B was selected as the final 10-class model. Evaluated on a held-out 500-row stratified test sample for the first time, Model B achieved accuracy of 0.58, macro-F1 of 0.5761, and ±1 accuracy of 83.6%, closely consistent with validation performance, confirming the model generalizes well to unseen data.

3-class modeling showed that re-bucketing Model B's existing 10-class predictions into three classes already produces strong results; accuracy 0.89, macro-F1 0.80 on the full validation set, since most 10-class errors are near-misses that collapse within the same bucket. On the 500-row test sample, the 3-class performance reached accuracy 0.91 and macro-F1 0.84, with Class 3 performing strongest (F1 0.94) and Class 2 remaining the most challenging (F1 0.67), consistent with its minority status in the dataset. A dedicated CNN trained from scratch on the 3-class task achieved macro-F1 of 0.7062 on the full validation set; below the re-bucketed DistilBERT baseline, though neither model reached convergence and both results should be read as lower bounds.

Explainability confirmed that both DistilBERT and CNN models reason in interpretable, clinically meaningful ways. LIME analyses showed high confidence at rating extremes (near-100% for ratings 1 and 10) and genuine uncertainty at middle ratings, tracking real linguistic ambiguity in the reviews. A global leave-one-out analysis identified consistent word-direction patterns across both architectures: strongly positive outcome words ("saved," "miracle," "love") push toward high ratings, while side-effect and failure language ("bleeding," "cramps," "suffered") push toward low ratings. SHAP cross-validation on the CNN confirmed LIME's findings using an independent attribution method.

Generative AI augmented the pipeline with Qwen3-4B, producing ≤10-word summaries and binary sentiment labels for a 500-row test sample. Four prompting approaches were tested, progressing from zero-shot to clinically descriptive few-shot with role framing. Approaches B and D were selected as the two final outputs, both achieving $\sim$77% sentiment accuracy. The analysis revealed a meaningful tradeoff: B abstains more readily (57 unknowns, higher precision), while D commits more confidently at the cost of occasional mislabeling (8 unknowns, lower precision on negative). Both approaches agree on 83.2% of reviews, with disagreements concentrated in genuinely ambiguous cases, most notably reviews describing mixed clinical experiences that resist clean binary classification.

Across all components, the most consistent finding is that the inherent ambiguity of middle-range patient experiences (ratings 4–7, Class 2) is the primary challenge; for the classification models, the GenAI model, and the sentiment framing alike. Addressing this, through more training, larger evaluation samples, a dedicated 3-class DistilBERT, and a 3-class sentiment framing in the GenAI component, represents the clearest path forward identified in this work.

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